



Hack-Nation

In collaboration with MIT Club of Northern California and MIT Club of Germany

CHALLENGE

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The Negotiator

Voice agents that call, compare, and haggle — pick your market, never overpay again

POWERED BY

|| ElevenLabs

ElevenLabs

(The AI Audio and Voice Agents Platform)

Goals and Motivation

Meet Daniel

Daniel is 34, has just signed a lease in Charlotte, and is sitting on the floor of his apartment in Rock Hill, 45 miles away, on a Sunday evening — three weeks to moving day, a two-bedroom flat around him, phone in hand.

The market knows quite a lot about what this move should cost. Real quotes collected for exactly this move — Rock Hill to Charlotte, 45 miles, standard service — ranged from \$1,158 to \$6,506. A 5.6x spread for identical work. Somewhere in that range is a fair price, and the only way to find it is to call five to eight small moving companies, describe the same apartment in detail each time, sit through the hold music, compare fee structures that are deliberately hard to compare, and negotiate.

Almost nobody does this.

So Daniel does what most people do: he calls two companies, takes a sight-unseen phone quote from the friendlier one, and hopes. FMCSA data shows how that ends — estimates given without seeing the shipment are 40% more likely to produce a final bill above the original quote. On moving day, the stairs cost extra. So does the “long carry”. The price on the truck is not the price on the phone.

This is not an information problem in the classic sense. The prices exist. The red flags are documented. Industry guidance even tells consumers exactly what to do: gather multiple quotes, and treat anything 30% below the competition as a warning sign. What is missing is anyone with the time and stamina to extract those numbers, one phone call at a time.

That gap — between the price Daniel will pay and the price he could have paid — is what this challenge asks you to close. Not just for moving: for whichever phone-priced market you choose, B2C or B2B — medical bills, car buying, contractor bids, freight, equipment rental. For the first time, software can literally pick up the phone.

Current challenges

Every phone-priced market shows the same pattern; moving is simply the best documented. A \$20B+, deeply fragmented US industry — 16,851 companies averaging just 6.2 employees, serving roughly 28 million Americans who move every year — made of phone-and-paper operations, not tech businesses with instant online quotes. Swap in car repair, medical billing, or subcontractor bids and the same three failures repeat:

OPAQUE PRICING	Quotes for the same 45-mile move range from \$1,158 to \$6,506 — a 5.6x spread for identical work. The only sanity check available to a consumer is gathering multiple quotes, and there is no efficient way to do it today.
UNRELIABLE ESTIMATES	Sight-unseen estimates — which is what most phone quotes are — are 40% more likely to end in a final bill above the original quote (FMCSA). Lowball-then-upcharge is a business model, not an accident.
NO WAY TO SHOP AROUND	The market is 16,851 small companies reachable one phone call at a time. The BBB logs 13,000+ complaints against movers every year, largely over unexpected price hikes — yet almost nobody has the hours to call 5–8 operators and compare properly.

Your Challenge

Build THE NEGOTIATOR — a working end-to-end MVP of a voice-agent system that, for a vertical of your choice, gathers real prices by phone, reports them in comparable form, and negotiates the best deal. B2C or B2B is up to you: moving, medical bills, car buying, contractor bids, freight, equipment rental, wedding vendors — pick the market where you see the biggest need and can prove the pain with real numbers.

Whatever the vertical, the system always has the same three beats. First, intake: build a complete, structured specification of the job — through a voice interview, through documents the user already has (photos, existing quotes, bills, inventory lists), or both. Second, the calls: your agent phones the market — whether real businesses, humans role-playing, or counter-agents you build — describes the job identically every time, and extracts itemised quotes. Third,

the close: it negotiates — leveraging one bid against another, questioning fees, flagging suspicious outliers — and reports back with a ranked, evidence-backed comparison: call recordings, transcripts, and a recommended deal explained in plain language. Price gathering, reporting, and negotiation are all mandatory; the vertical is yours.

Conversation design is as much part of the challenge as the technology. How your agent handles a gruff dispatcher, a “we don’t give prices over the phone”, or a hard sell determines whether it comes back with a usable quote or a useless range.

A negotiator — not a quote form

Your solution must conduct real voice conversations, not fill web forms or scrape aggregators. The other end of the line is flexible: bring a real problem you are facing right now and let the agent make real calls; answer the calls yourself, playing different counterparts — the tough negotiator, the stonewaller, the upseller; or build counterparty agents and run agent-to-agent. All are fine, as long as you can simulate several distinct negotiation types and prices move during the call because of what your agent knows and says. Vertical-specific parameters (job-spec taxonomy, price benchmarks, red-flag rules, negotiation levers) should be configuration, not code: switching your system from movers to auto body shops should mean swapping a config file, not rewriting your agents.

Build all three of these modules:

01 The Estimator — Intake by Interview or Documents

An intake agent builds a complete, structured job specification — the thing that makes a later quote binding rather than bait. Two intake paths, and strong submissions offer both: a voice interview that asks what a professional estimator would ask, and document intake — photos, existing quotes, bills, inventory lists — parsed into the same structured spec. In moving that spec is rooms, large items, and stairs; in medical billing it is the itemised bill; in construction it is the scope of work. The user confirms the spec before any calls are made. This is the direct attack on the sight-unseen problem: incomplete intakes are why estimates blow up 40% of the time.

Required: *Intake must offer a voice interview built on ElevenLabs Agents plus at least one document type; both paths must produce the same structured job spec (e.g. JSON), confirmed by the user and reused verbatim across every call.*

02 The Caller — Parallel Quote Gathering

Your agent phones the market. How the other end answers is up to you — three setups are all valid: (1) real calls — point the agent at a problem you are actually facing this month and let it phone real businesses via Twilio/SIP; (2) human-in-the-loop — answer the calls yourself, playing distinct counterparts: the tough negotiator, the lowballer with hidden fees, the hard-sell upseller; (3) simulated market — build counterparty agents and run agent-to-agent. Whatever the setup, the caller must describe the job consistently, handle friction (interruptions, evasive answers, “someone will call you back”), and extract an itemised, comparable quote from each call. Use batch calling or parallel sessions where you can — and show where the call list would come from in the real world (e.g. Google Places or Yelp business data).

Required: Any counterparty setup is allowed — real businesses, humans role-playing, or built counter-agents — but your demo must show live calls against at least three distinct negotiation styles, every quote captured in structured, comparable form with fees itemised.

03 The Closer — Negotiation & Reporting

With quotes in hand, your agent negotiates: it leverages competing bids (“I have a binding quote for \$1,850 — can you beat it?”), pushes on fees, asks for price-matching or extras, and applies red-flag rules — any quote 30%+ below market is a warning sign, not a win. Then it reports: a ranked comparison the user can trust, with a recommended deal, full transcripts and recordings, itemised fee breakdowns, and a plain-language explanation of why.

Required: At least one demonstrated negotiation in which the price or terms change during the call because of leverage your agent gathered; the final report must rank all quotes and cite transcript evidence.

The Conversation Requirement

Voice is not a skin on a chatbot; it is the mechanism of trust on both ends of the call. Your prototype must address the following explicitly:

- Who is the agent speaking for? Does it disclose that it is an AI calling on behalf of a customer? How does it respond to “am I talking to a robot?” — gracefully, honestly, and without losing the quote?
- How does it survive friction? Busy dispatchers interrupt, answer vaguely, and multitask. Latency, barge-in handling, and turn-taking decide whether the call sounds like a serious buyer or a bot.
- Where does the honesty line sit? The agent may use competing quotes as leverage — but it must never invent inventory, fake a bid, or misrepresent the job. Show how you constrain it.
- How does every call end? With a structured outcome: an itemised quote, a callback commitment, or a documented decline — never a vague “they said around two thousand”.

Play the calls in your demo and highlight how you addressed these four points.

Data Sources and Hints

Your agents should be grounded in real market data and live voice infrastructure — not canned audio. You are encouraged to build on the resources below, adapted to the vertical you choose.

VOICE AGENTS & TELEPHONY	
ElevenLabs Agents Platform	Build and orchestrate the conversational agents — system prompts, tools, knowledge bases, agent transfer, and human handoff. (elevenlabs.io/docs)
Batch Calling / Twilio & SIP	Run parallel outbound calls at scale; native Twilio integration and SIP trunking make real phone calls possible if you want to go beyond simulation.
Agent Tools & MCP	Give your agents tools to log structured quotes mid-call, look up price benchmarks, and write results into your comparison backend.
MARKET DISCOVERY & PRICING DATA	
Google Places / Yelp Fusion / OSM	Find the counterparties: business phone numbers, categories, opening hours, ratings, and review volume by location — your call list, built programmatically instead of by hand.
Vertical price benchmarks	Ground truth for red-flag detection, per vertical — e.g. FMCSA & moveBuddha (moving), FAIR Health & hospital price-transparency files (medical), RepairPal (auto repair), KBB (car buying), published rate cards (freight, equipment rental).
Document parsing	Turn what the user already has — photos, existing quotes, bills, inventory PDFs — into the structured job spec, using vision models or OCR. Same schema as the interview output.
THE OTHER END OF THE LINE	
Counterparty options	Three valid setups, any mix: real calls to a problem you actually face right now; yourself on the line role-playing distinct counterparts (tough negotiator, stonewaller, upseller); or built counter-agents with different pricing strategies. Cover several negotiation types.
Golden calls & eval sets	Record reference negotiations and build simple evals: does the agent extract every fee? Does it catch the 30%-below-market red flag? Use these to iterate on your prompts.

What Makes a Strong Submission

Strong Submissions...	Weak Submissions...
— Show a real negotiation. Demonstrate a price moving during a live call — against a real business, a role-playing human, or a counter-agent — because of leverage your agent gathered, not because the script said so.	— Stage a screenplay. Two agents reading pre-written dialogue at each other is a demo of text-to-speech, not of negotiation.

— Pick a provable pain. Choose a vertical where you can show the spread with real numbers — then make quotes comparable: same job, same services, itemised fees.	— Pick a vertical because it sounds cool, collect prices that aren't comparable, and hide the differences in a polished dashboard.
— Close the loop. Intake → calls → negotiation → ranked recommendation with transcript evidence. A connected flow beats a polished fragment.	— Over-engineer the agent stack and under-engineer the conversations. This challenge is won in call design, not model architecture.
— Be honest about the hard parts. AI disclosure, honesty constraints, and failure modes (hang-ups, refusals to quote) handled gracefully build trust.	— Let the agent bluff. Invented inventory or a fake competing bid may win \$200 in the demo — and lose the user, and the market, forever.

Success Criteria

A submission fully meets this challenge when:

- The loop is closed. A single vertical runs end to end: intake → calls → negotiation → ranked recommendation with transcript evidence.
- One structured job spec, built by voice interview and at least one document type, is confirmed by the user and reused verbatim across every call.
- Live calls are demonstrated against at least three distinct negotiation styles; every quote is captured in structured, comparable form with fees itemised.
- At least one negotiation shows the price or terms measurably change during the call because of leverage the agent gathered — not because the script said so.
- AI disclosure and honesty constraints hold: the agent never invents inventory or a fake bid, discloses it is an AI when asked, and handles friction (hang-ups, refusals, “are you a robot?”) gracefully.
- Every call ends in a structured outcome — an itemised quote, a callback commitment, or a documented decline.
- The final report ranks all quotes, cites recordings and transcripts, and explains the recommended deal in plain language.

Why This Matters

The consumer harm is documented, not hypothetical: the BBB logs over 13,000 complaints and negative reviews against movers every year, largely over unexpected price hikes; “hostage load” cases — belongings held until the customer pays more — have nearly tripled since 2017; final bills are routinely inflated 30%+ through added fees; and moving prices have risen roughly 21% recently on fuel costs alone. Consumers have more reason than ever to shop around, and no time in which to do it.

For ElevenLabs, this is the thesis in miniature: a \$20B+ market that runs on the telephone, where 16,851 small operators will never adopt quoting software — but every one of them answers the phone. A voice agent is the first technology that meets this market where it is, with no integration required on the other end. And moving is only the beachhead: the same system generalises to contractors, auto repair, storage, and medical bills — every market where the real price is only available by phone, to whoever has the patience to ask for it.